Low-cost soil moisture sensing for smallholder irrigation scheduling: a two-season field trial

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Background

Smallholder vegetable growers in the savanna zone irrigate by habit rather than by measurement. Over-irrigation wastes pumped water and fuel, while under-irrigation during fruit set cuts yields sharply.

Commercial tensiometers cost more than a season of fuel, so we tested whether a capacitive sensor costing under five dollars could guide irrigation decisions.

Objectives

- Calibrate a low-cost capacitive probe against gravimetric soil samples.
- Compare sensor-guided scheduling with farmers' usual practice.
- Measure water use, onion yield and net income per plot.

Methods

Twenty-four farmers in four communities each managed two adjacent onion plots of 200 square metres over the 2025 and 2026 dry seasons. One plot followed usual practice and the other was irrigated only when the sensor reading fell below a threshold set at 55 percent of available water.

Raw sensor counts were converted to volumetric water content with a site-specific linear model:

```
import numpy as np

# raw ADC counts vs. gravimetric VWC (m3/m3), 36 samples
slope, intercept = np.polyfit(counts, vwc, 1)

def to_vwc(raw):
    return max(0.0, slope * raw + intercept)

THRESHOLD = 0.55 * (FC - PWP) + PWP
```

The calibration explained 91 percent of the variance in measured water content (RMSE 0.021 m3/m3).

Results

Table 1. Mean values per plot over two seasons (n = 24 farmers)

Indicator	Usual practice	Sensor-guided	Change
Irrigation events per season	41	27	−34 %
Water applied (mm)	612	438	−28 %
Pump fuel (litres)	96	69	−28 %
Bulb yield (t/ha)	18.4	19.7	+7 %
Net income (GHS per plot)	1 520	1 790	+18 %

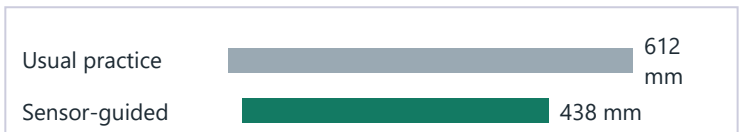


Figure 1. Seasonal water applied, mean of both seasons.

Sensor-guided plots received 28 percent less water while yielding slightly more onions. The saving was largest in the first six weeks after transplanting, when farmers usually irrigate every second day.

Three sensors failed after flooding of the field channel and were replaced within a week.

Discussion

Farmers trusted the readings once they had seen the probe respond to a single irrigation. Several asked for a simple display of three coloured lights instead of a number.

The yield gain is modest, but the fuel saving alone repays the sensor within the first month.

Conclusions

- A five-dollar capacitive probe, calibrated once per soil type, is accurate enough to schedule irrigation.
- Sensor-guided scheduling saved over a quarter of the water and fuel without any loss of yield.
- Adoption depends on a display that farmers can read at a glance.

Next steps

In 2027 the trial will extend to tomatoes and peppers in two further districts, with the three-light display built by a local electronics cooperative.

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References

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